

Instructions for the experimenter

In `abm_agent_H.py`, adapt the PsychoPy monitor size parameter to the monitor size of the device. In `abm_experiment.py`, adapt the participant ID, number of blocks, and number trials per block parameters

Participant instructions

Gabor patch contrast instructions

On each trial of the task, you will see two Gabor patches to the left and right of a fixation cross. One of the patches has a higher contrast, i.e. a larger difference between the black and the white parts of the image, than the other. Your task is to indicate by pressing the left or right cursor which of the Gabor patches has the higher contrast. If your answer is correct, you receive a reward of +1, if your answer is incorrect, you receive a reward of +0. Your aim is to maximize your sum of rewards, i.e. to be correct as often as possible. Sometimes, the difference in contrast between the Gabor patches is obvious, sometimes it is more subtle. After you have been familiarized with the task, you will perform 10 blocks of 30 trials each.

Symmetric bandit instructions

On each trial of the task, you can choose between pressing the left or the right cursor button. For a given block of the task, each button is associated with fixed probabilities that pressing it will yield a reward of +1 or +0. For example, on a given block, pressing the left cursor button will yield a reward of +1 with probability 0.8 and will yield a reward of +0 with a probability of 0.2. When pressing the left button, you will thus observe a +1 reward on average in 8 of 10 trials and a reward of +0 in 2 of 10 trials. Pressing the other button (in this example, the right button) will yield the rewards with opposite probabilities assigned to them. In the example, if pressing the left button yields a reward of +1 with probability 0.8, then pressing the right button yields a reward with probability 0.2, and likewise for obtaining rewards of +0. One of the buttons is thus always associated with a higher probability of obtaining a reward of +1 than the other button. Sometimes this difference is pronounced, sometimes it is more subtle. Your aim is to maximize your sum of rewards, i.e., to press the button with the higher associated probability for +1 as often as possible. Which button that is, you must figure out on each block, as the probabilities will be reassigned once a new block begins. After you have been familiarized with the task, you will perform 10 blocks of 30 trials each.